INCH-POUND

MIL-C-3098/80D
28 June 1995
SUPERSEDING
MIL-C-3098/80C
2 November 1983

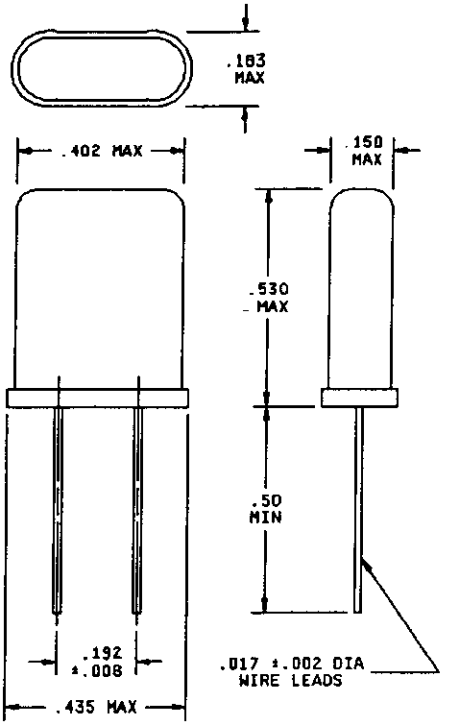
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR105/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirement for the acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-3098.

Pertinent characteristics: 48 to 125 MHz; fifth mechanical overtone; noncontrolled; series resonance.



Inches	mm
.002	0.05
.008	0.20
.017	0.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.74
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-C-3098.

FIGURE 1. Crystal unit - CR105/U.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS

Holder: HC-18/U or HC-43/U (MIL-H-10056/9) or HC-49/U (MIL-H-10056/36).

Dimensions, marking, and configuration: See figure 1.

Frequency range: 48.0 to 125.0 MHz, inclusive.

Capacitance, shunt: 4.0 pF, maximum.

Mode of oscillation: Fifth mechanical overtone.

Rated drive level: 2.0 ±0.4 mW.

Equivalent resistance: 40 ohms.

Resonance: Series.

Reference temperature: +25°C ±1°C.

Frequency tolerance: ±10 ppm.

Operating temperature range: -40°C to +95°C, inclusive.

Frequency tolerance: ±25 ppm.

Equivalent resistance: 60 ohms, maximum.

Unwanted modes: Method II, minimum unwanted-to-main mode resistance ratio of 3:1.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change: ±10 percent.

Vibration: Method 204 of MIL-STD-202, test condition A.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Thermal shock:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±10 percent.

Aging:

Frequency change permitted: ±5 ppm.

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

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